

1 Theoretical Framework

To establish our fault-tolerant state preparation protocol, we must bridge the gap between the abstract stabilizer formalism of CSS codes and the physical operations required for fault-tolerant circuit extraction. We utilize the ZX-calculus as the intermediate graphical language, allowing us to topologically manipulate state preparation procedures before compiling them into quantum circuits.

1.1 ZX-Calculus Representation of CSS Codes

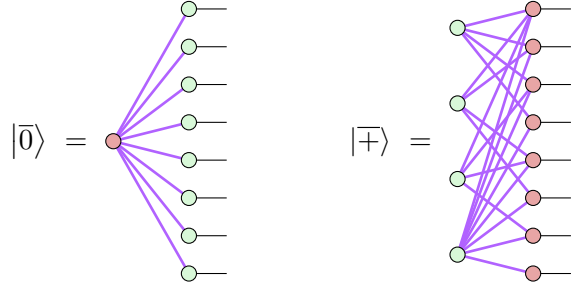
Any Calderbank-Shor-Steane (CSS) quantum error-correcting code can be natively represented as a ZX-diagram [?]. This representation is constructed by mapping the generators of the stabilizer group and the logical operators of the code into a network of interconnected spiders.

We will use the $[[8, 3, 2]]$ “smallest interesting color code” as a running example, whose stabilizer generators can be explicitly written as:

$$X_1 X_2 X_3 X_4 X_5 X_6 X_7 X_8 \quad (1)$$

$$Z_1 Z_2 Z_3 Z_4 \quad Z_1 Z_2 Z_5 Z_6 \quad Z_1 Z_3 Z_5 Z_7 \quad Z_1 Z_2 Z_3 Z_4 Z_5 Z_6 Z_7 Z_8 \quad (2)$$

From these stabilizers, one can construct encoders for the code or the fault-tolerant state preparation ZX-diagrams:

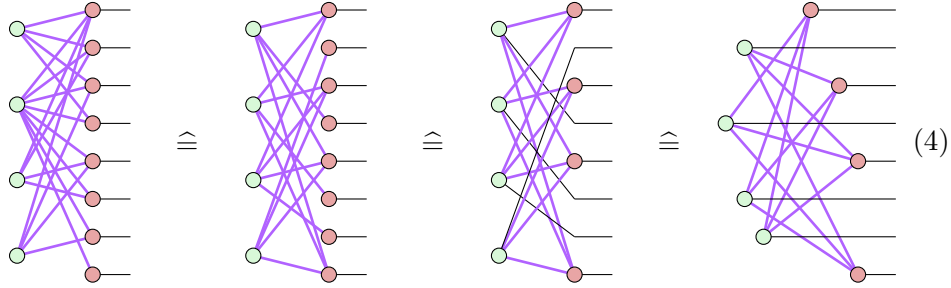


$$|\bar{0}\rangle = \quad |\bar{+}\rangle = \quad (3)$$

In this representation, the correlations between stabilizers are denoted by “perfect” edges. As established in Section ??, a perfect edge serves as a mathematical specification of an idealized, noiseless logical correlation. Because physical hardware cannot natively execute noiseless operations without an infinite distance code, these edges serve only as an intermediate topological tool. They must be systematically eliminated from the diagram to extract a valid, physically realizable fault-tolerant circuit.

1.2 Bipartite Structural Constraints and Systematic Form

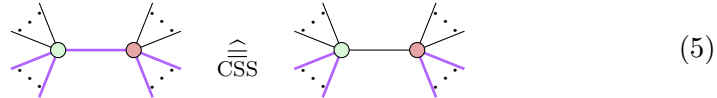
[TODO: Move the theoretical requirement for Systematic Form here. Explain that an arbitrary parity check matrix M does not cleanly decompose. Prove/state that if M contains a full-rank identity submatrix (systematic form), the resulting ZX diagram is strictly bipartite, granting each spider a dedicated physical output leg. This structural guarantee is required for the subsequent rewrites.]



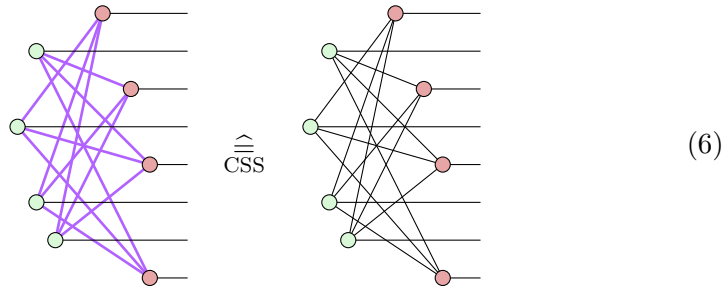
1.3 Fault-Equivalent Edge Elimination

To extract a physically realizable quantum circuit, we must find a fault-equivalent ZX-diagram with no perfect edges. To achieve this, we introduce the following fault-equivalent (FE) ZX-rewrite rule:

Proposition 1.

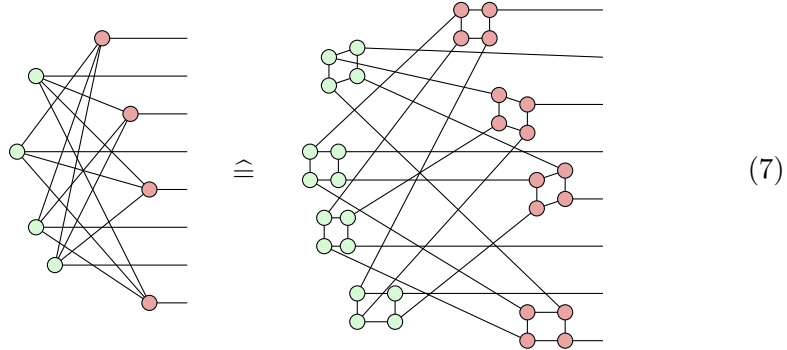


[TODO: Formally state the condition: A perfect edge can be eliminated iff both spiders possess a physical leg. Note that because we mathematically restricted the matrix to a bipartite graph state in the previous section, this condition is globally satisfied.]

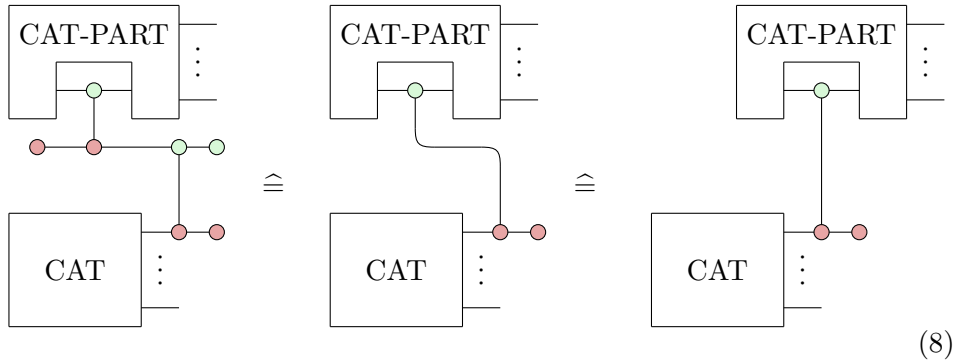


1.4 Spider Decompositions and Measurement-Free Connectivity

Once the bipartite ZX-diagram is free of perfect edges, the remaining graph consists of high-degree physical spiders. Directly mapping high-degree spiders to physical hardware violates native 2-qubit gate constraints. Therefore, each high-degree spider must be fault-equivalently decomposed into a network of degree-3 spiders utilizing the SpiderCat protocol [TODO: Add citation].

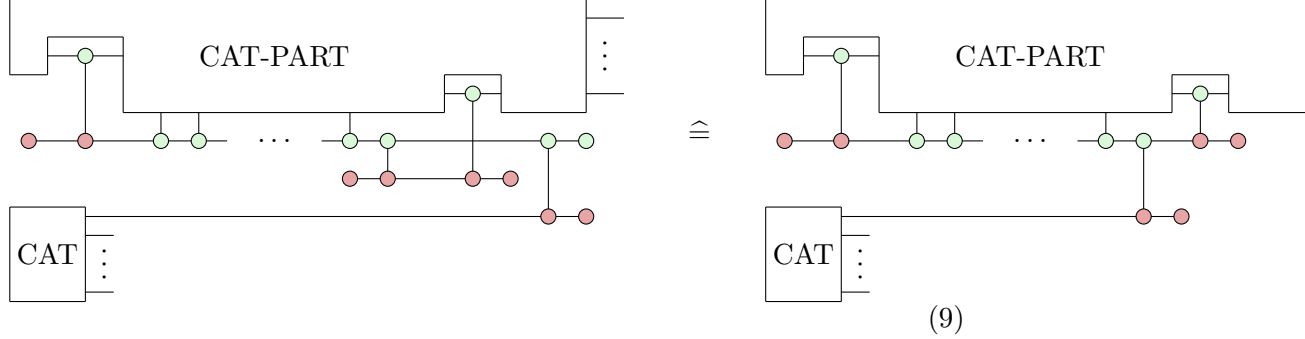


A central challenge in extracting the overall CSS state preparation circuit is connecting these individually prepared cat states without introducing prohibitive Bell-state measurement overhead. While previous protocols (e.g., *flag-at-origin*) resolve this for unbranched trees by substituting the Bell measurement with a direct CNOT connection (Eq. 8), standard SpiderCat introduces internal branching, rendering that specific rewrite inapplicable.



To preserve the depth advantages of branching while avoiding measure-

ment overhead, we introduce a novel flag-delegated connection rewrite:



This topological substitution allows a designated flag qubit to absorb the classical measurement, connecting the independent cat states with identical resource savings. We formalize the applicability of these combined rewrites with the following proposition:

Proposition 2. Consider a degree- n spider decomposed into a physical circuit via a spanning forest. If all but one of its boundary legs satisfy either the direct CNOT condition (a single state preparation followed by a single CNOT) or the flag-delegated condition (the final CNOT acting on the boundary qubit is the initial CNOT acting on a flag qubit), then the entire spider can be directly connected to the broader CSS state without requiring intermediate Bell-state measurements.

While standard SpiderCat minimizes gate counts, it does not guarantee the strict spanning forest topologies required by Proposition 2.

2 The Compilation Pipeline

To physicalize the theoretical constructs outlined in Section 1, we introduce a comprehensive compilation pipeline. This algorithm accepts an arbitrary CSS code parity check matrix and deterministically outputs a highly optimized, fault-tolerant state preparation circuit. The pipeline operates in three distinct phases: (1) logical resource optimization to find the ideal bipartite basis, (2) topologically constrained spider generation to satisfy Proposition 2, and (3) acyclic global edge routing to synthesize the final circuit.

2.1 Phase 1: Resource Characterization and Basis Optimization

Our pipeline begins by quantifying the logical resources required to prepare a given state. Given an X or Z spider of degree n , we characterize the exact logical CNOT cost assuming a maximum tolerance of t faults.

The density lower bound, $\delta(t)$, governs the necessary graph density:

$$\delta(t) = \frac{\lceil (t+3)/2 \rceil \lfloor (t+3)/2 \rfloor}{\lceil (t+3)/2 \rceil \lfloor (t+3)/2 \rfloor + \lceil (t-3)/2 \rceil \lfloor (t+3)/2 \rfloor + \lfloor (t-3)/2 \rfloor \lceil (t+3)/2 \rceil} \quad (10)$$

Because a degree- n spider represents a cat state stabilized by a global parity operator, the maximum number of faults requiring correction is bounded: $t' = \min(t, \lfloor n/2 \rfloor - 1)$. By the handshaking lemma for 3-regular graphs, the final edge count E and vertex count V are determined by:

$$E = 3 \left\lceil \frac{\lfloor n/\delta(t') \rfloor}{3} \right\rceil, \quad V = \frac{2E}{3} \quad (11)$$

The minimum number of flag qubits is $f(n, t) = \max(0, \lceil E - V + 2 \rceil - 1)$. The logical CNOT cost for a single isolated spider is bounded by: $C_{\text{spider}}(n, t) = n - 1 + 2f(n, t)$.

Row Optimization via Stochastic Basis Search

For the state preparation circuit to decompose cleanly, the corresponding ZX diagram must represent a bipartite graph state (Section 1). Thus, the parity matrix M must assume systematic form. The total logical CNOT cost of the resulting circuit is:

$$C_{\text{total}}(M, t) = \sum_{\substack{c=1 \\ w_c > 1}}^c 2f(w_c + 1, t) + \sum_{r=1}^r [w_r - 1 + 2f(w_r, t)] \quad (12)$$

where w_c and w_r are the column and row Hamming weights.

Because finding the globally optimal basis over $\mathbb{F}_2$ is computationally prohibitive, Phase 1 employs a Monte Carlo optimization heuristic. We randomly sample subsets of r columns from M to form $r \times r$ submatrices. If a submatrix M_{sub} is non-singular, we compute the row-equivalent matrix $M_{\text{new}} = (M_{\text{sub}}^{-1} M) \pmod{2}$. We evaluate $C_{\text{total}}(M_{\text{new}}, t)$ for each valid basis and retain the matrix that minimizes the total logical CNOT cost.

2.2 Phase 2: Constrained Spider Decompositions via Simulated Annealing

[TODO: Draft Phase 2. Explain how the code translates individual spiders into physical structures.]

- Explain the necessity of the Minimum Diameter Spanning Forest (MDSF) to minimize circuit depth (longest root-to-leaf path).
- Explain the Simulated Annealing heuristic used to mutate the forest while preserving k -components.
- Crucially, detail the *topological penalties* applied during generation (e.g., degree locking and penalizing leaf adjacency) that actively force the algorithm to satisfy Proposition 2. This is what transforms standard SpiderCat into your "Modified" version.

Dependency Graph Construction and Acyclic Resolution

The dependency graph D is constructed via a two-step topological mapping:

1. **Tree Traversal Edges:** For every edge in the spanning forest F , we add a directed edge in D originating from the root and flowing outward towards the leaves. This represents the sequential propagation of the cat state via CNOT gates.
2. **Connection Scheduling Edges:** To guarantee the applicability of the flag-delegated rewrite (Eq. 9), we must constrain the extraction order of the circuit. For every leaf node l in F , we identify its neighbors in the original graph G that are not connected via F . We append a directed edge from l to each of these missing neighbors in D . By enforcing this dependency, the subsequent topological sort ensures that the final CNOT gate acting on a data qubit will deterministically be the initial CNOT gate of its corresponding flag.

By enforcing these dependencies, the graph D strictly models the causal requirements of our connection rewrites. However, because the underlying physical connectivity contains closed spatial loops, these temporal constraints inevitably introduce directed cycles into D . A directed cycle represents a temporal paradox in quantum circuit compilation: a sequence of operations must strictly precede itself, rendering extraction impossible.

To resolve this, we exploit the single degree of freedom permitted by Proposition 2. The proposition dictates that exactly one boundary leg is

exempt from the strict flag-delegated conditions, as it must serve as the dedicated logical output connecting the spider to the broader bipartite graph state.

Algorithmically, this translates to removing exactly one connection scheduling edge from D . Our modified SpiderCat implementation systematically iterates through all such edges, temporarily removing each to test the topology of the remaining graph. We search for a single scheduling edge whose removal collapses all directed cycles, transforming D into a strict Directed Acyclic Graph (DAG).

The removal of this specific edge designates the physical qubit that will serve as the logical output of the cat state. The resulting DAG inherently defines a valid partial ordering (a topological sort). Traversing this DAG guarantees a deterministically ordered, paradox-free quantum circuit that respects all spatial constraints and natively eliminates Bell-state measurement overhead.

2.3 Phase 3: Acyclic Edge Routing and Global Synthesis

With the individual spiders decomposed into highly optimized, topologically constrained spanning forests, the next step is connecting these isolated components into a unified bipartite graph state.

A naive assignment of logical edges to physical nodes can inadvertently introduce temporal cycles—situations where a sequence of CNOT gates creates a closed causal loop, rendering extraction impossible. To strictly prevent this, Phase 3 employs a Minimum Remaining Values (MRV) backtracking constraint solver that simultaneously satisfies both the spatial connectivity constraints of the physical graph G_{global} and the temporal causality constraints of the global dependency graph D_{global} .

[TODO: The backtracking solver actively prunes branches that create temporal cycles. However, add a formal statement clarifying the topological conditions under which this solver is mathematically guaranteed to succeed. If it is purely heuristic and might fail on certain topologies (requiring a fallback to Phase 1 for a new matrix basis), state this explicitly to ensure rigorous accuracy.]

The solver maps the required logical connections dictated by the parity check matrix M' to specific physical nodes across the constituent Z and X spiders. For every proposed physical CNOT connection between a candidate Z-spider node and an X-spider node, the algorithm evaluates the dual-graph implications:

1. **Spatial Routing (G_{global}):** The algorithm provisionally assigns a physical, undirected CNOT edge in the global physical graph.
2. **Temporal Routing (D_{global}):** To preserve causal integrity, the algorithm identifies the strict temporal predecessors of the matched nodes from their respective local digraphs. It then provisionally injects directed cross-edges into D_{global} , linking the predecessors of the Z-node to the X-node, and vice versa.

If the injected directed edges introduce a cycle in D_{global} , the branch is aggressively pruned, and the solver rolls back the state to test the next valid candidate pair. If successful, this algorithm outputs a global graph G_{global} , a unified spanning forest F_{global} , and a strictly acyclic dependency graph D_{global} representing the exact temporal partial ordering required for valid circuit extraction.

2.4 Phase 4: Temporal Circuit Extraction

The final phase of the pipeline compiles the global topological structures ($G_{\text{global}}, F_{\text{global}}, D_{\text{global}}$) into a physically realizable, fault-tolerant quantum circuit. We execute this via a topologically sorted Breadth-First Search (BFS) traversal over the acyclic dependency graph D_{global} .

[TODO: Explain the dynamic qubit allocation logic. Specifically, briefly detail how the BFS utilizes your depths and branch_mark.values heuristics to decide when a newly traversed node inherits the primary data qubit, when it spawns a secondary data qubit, or when it initializes a dedicated flag qubit.]

The extraction procedure operates layer-by-layer according to the topological generations of D_{global} , ensuring no operation violates causality. The core mechanisms of the extraction are as follows:

1. **Initialization and Propagation:** Root nodes of the spanning forest allocate new physical data qubits initialized in the appropriate basis ($|+\rangle$ for Z-spiders, $|0\rangle$ for X-spiders). As the BFS traverses tree edges in F_{global} , CNOT gates are appended to propagate the state.
2. **Inter-Spider Connections and Flag Measurement:** When the traversal encounters a non-tree edge in G_{global} connecting two identical spider types, it triggers the flag-delegated rewrite (Eq. 9). The algorithm dynamically allocates a flag qubit, applies the requisite CNOT, and immediately performs the appropriate basis measurement to finalize the internal connection.

3. **Cross-Tree Linking and Parity Detection:** Inter-spider edges (Z to X) are compiled as direct physical CNOT gates. Intermediate parity measurements derived from the flag qubits and cross-tree boundaries are explicitly annotated as logical detectors to facilitate fault-tolerant syndrome analysis.

The output of Phase 4 is a complete, hardware-agnostic circuit (e.g., exported natively as a Stim circuit) consisting strictly of initializations, 2-qubit CNOT gates, measurements, and deterministic detector annotations, ready for physical hardware deployment.